Conversational Clustering

Can natural-language feedback steer clustering toward the user's intent — even when the embedding's default similarity disagrees?

THE PROBLEM

Clustering is ill-posed without knowing the user's intent. Same 200 abstracts admit many valid clusterings: topic, methodology, object scale, writing register.

THE STATUS QUO

Users tune K, distance metrics, and features by hand — iterating on knobs they don't understand. Most don't know what they want until they see what they didn't want.

THE PROPOSAL

Let the user steer the clustering through natural-language feedback turn-by-turn. The LLM maps feedback to operations on a classical pipeline.

Research Questions

What we are actually trying to measure

HEADLINE

Does conversational refinement converge to a hidden target clustering more efficiently than one-shot prompting — and does this advantage depend on whether the target aligns with or conflicts with the embedding's default similarity structure?

RQ1

Efficiency

Does conversation reach a target ARI threshold in fewer turns than iterated one-shot?

RQ2

Ceiling

Does conversation reach higher final ARI against the hidden target?

RQ3

Bias override (the interesting one)

How does the gap shift as targets move from aligned with embedding defaults (topic) to conflicting (methodology, object scale)?

RQ4

Stability

Is the converged clustering stable across runs and corpus resamples?

Scope *· solo · arXiv astro-ph abstracts (~2000) · simulated-user evaluation with hidden targets · 3 targets × 2 systems × ~15 runs each*

Where this sits in the literature

LLM-guided clustering is crowded — the measurement angle isn't

Closest prior art

- **ClusterLLM (Zhang et al., EMNLP 2023)**

  User-specified 'perspective' + LLM-judged triplets refine embeddings.

- **ITGC (Iterative Text-Guided Clustering, 2025)**

  Iterative refinement guided by natural-language instructions — on images.

- **InBedder (Peng et al., 2024)**

  Instruction-following embeddings — change what's encoded to change clusters.

- **Dial-In LLM (2024)**

  LLM-in-the-loop intent clustering with cluster-level refinement.

- **Viswanathan et al. (2024)**

  Few-shot LLM-guided clustering at three pipeline stages.

THE GAP I'M AIMING AT

Most prior work evaluates against targets that align with embedding defaults.

Week 1 finding on astro-ph: sentence-transformer embeddings encode at least three axes simultaneously — topic, methodology, object scale. Any single user goal will align with one and conflict with the others.

The contribution isn't a new system. It's a measurement: how does interactive refinement perform when the user's target conflicts with what the embedding wants to cluster on?

Honest stance — system architecture is not novel relative to ClusterLLM / ITGC / InBedder. The contribution is empirical, not architectural.

Initial experiment design

Simulated-user methodology, three hidden targets of varying difficulty

1	**2**	**3**	**4**	**5**
Experimenter	**Simulated user**	**System**	**New clustering**	**ARI logged**
defines hidden target	knows target, gives natural-language feedback	feedback ⟶ ops on the pipeline	produced by applying operations	system output vs hidden target

↻ Loop steps 2–5 until convergence (Δ ARI < ε or turn budget exhausted)

Three hidden targets · varying difficulty

EASY

Topic

arXiv primary categories (GA / SR / CO / EP / HE / IM). Aligns with embedding default. Baseline should do well.

HARD

Object scale

{planetary, stellar, galactic, cosmological}. Hand-labeled. Embedding has partial signal.

HARDEST

Methodology

{observational, theoretical, simulation, instrumental}. Hand-labeled. Conflicts with embedding default.

Primary metric: ARI trajectory vs hidden target across turns. **Secondary:** turns-to-threshold, final-ARI variance, operation-mix descriptive analysis. **Stats:** bootstrap CIs, BH correction across the (3 targets × multi-metric) test family.

System architecture — initial sketch

LLM steers a classical clustering pipeline through a defined operation vocabulary

arXiv astro-ph corpus

~2000 abstracts via arXiv API · cached locally

Sentence-transformer embedder

all-MiniLM-L6-v2 · 384-dim, normalized

k-means clustering

current K, current cluster assignments

Cluster summarizer (LLM)

produces short per-cluster description for the user

→

LLM router

OPERATION VOCABULARY

Free-form feedback is parsed by an LLM into one or more typed operations on the pipeline.

`change_k(new_k)`

re-run k-means with a new cluster count

`split(cluster_id, criterion)`

sub-cluster a single cluster (LLM proposes the cut)

`merge(c_a, c_b)`

combine two clusters into one

`move(items, target)`

manually reassign specific items

`re-embed(emphasis)`

rewrite text emphasizing an axis, re-embed — the sharpest tool for bias override

`ignore(pattern)`

strip terms/phrases before embedding

Week-1 MVP: *split / merge / change_k only.* ***Stretch:*** *re-embed and ignore added once the loop runs end-to-end.*